

POLYMERIC ELECTROLYTES BASED ON SULFUR DIOXIDE COPOLYMERS

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(Received 20 December 1991)

Abstract—The present paper deals with applications of copolymers containing polar SO₂ units in main chains as matrices for polymeric electrolytes. Some of them exhibit conductivities exceeding 10⁻⁵ S cm⁻¹ at ambient temperatures. Blending of these copolymers with high molecular weight poly (ethylene oxide) results in an improvement of conductivities as well as mechanical and temperature stability of the studied electrolytes.

Key words: sulfur dioxide copolymers, polymer blends, polymer electrolytes, conductivity, differential scanning calorimetry.

INTRODUCTION

Sulfur dioxide participates in a wide variety of free radical copolymerizations with monomers having unsaturated carbon-carbon bonds to form polysulfones[1, 2] and in ionic copolymerizations with oxiranes to form polysulfites[2-4]. Since it is well known that compounds containing sulfur-oxygen bonds usually exhibit good cation complexing properties one could expect that the presence of highly polar SO_2 monomeric units in the chain may provide favorable conditions for fast ion transport in some solid polymeric electrolytes.

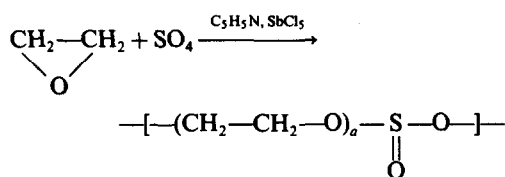
The aim of this paper is to describe preliminary conductivity data for several SO₂ copolymers doped with LiClO₄. Conductivities of blends of these electrolytes with high molecular weight PEO are also presented.

EXPERIMENTAL

The polysulfones were obtained in free radical copolymerization of sulfur dioxide with various vinyl monomers (allyl alcohol, vinyl acetate, vinyl oxvethylene ether, methyl acrylate and acrylamide).

The reactions were carried out in bulk using 10-fold excess of SO₂ at 195 K (I–IV) or 273 K (V) in the presence of t-butyl hydroperoxide as initiator according to the procedures described earlier^{5–7}.

The polysulfites were obtained by the ionic copolymerization of sulfur dioxide with ethylene oxide, carried out at room temperature in bulk using 10-fold excess of SO_2 in the presence of SbCl_3 (VII copolymer) or pyridine (VI copolymer) according to the procedures described elsewhere[2-4].



VI $a = 1$, VII $a = 1.8$

Compositions of the products were determined on the basis of elemental analysis and ^1H - or ^{13}C -NMR studies. The copolymers I, II, III and VI were found to be of an alternating structure. In copolymers IV and V sulfonic groups were separated by at least two acrylic monomeric units[5, 7]. The polysulfones were

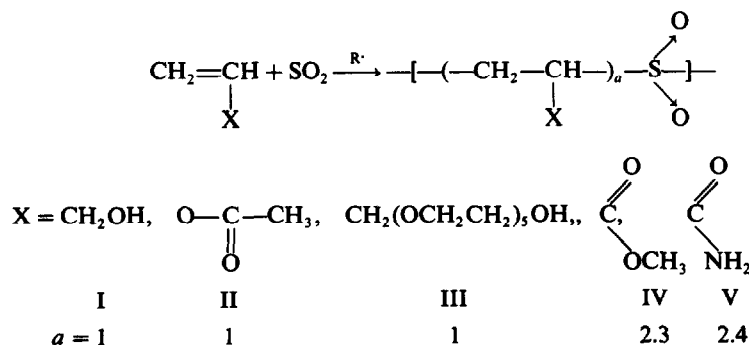


Table 1. Values of conductivity for electrolytes based on aliphatic polysulfones and their blends with PEO (samples doped with LiClO_4 10 mol% with respect to all monomeric units)

Kind of copolymer component	X/% wt	$\sigma_{\text{rt}}/\text{S cm}^{-1}$	T/K	$\sigma_T/\text{S cm}^{-1}$
PEO- LiClO_4 system	—	6.1×10^{-7}	363	2.0×10^{-3}
I	100	5.9×10^{-9}	374	4.6×10^{-7}
I + PEO	20	1.1×10^{-5}	369	3.3×10^{-3}
II	100	3.8×10^{-7}	373	5.3×10^{-6}
II + PEO	50	2.4×10^{-6}	376	4.1×10^{-5}
III	100	1.4×10^{-5}	319	7.6×10^{-5}
III + PEO	30	5.5×10^{-5}	313	1.4×10^{-4}
IV	100	1.2×10^{-7}	321	3.4×10^{-6}
V	100	1.1×10^{-5}	368	3.1×10^{-4}
V + PEO	40	3.2×10^{-5}	372	3.2×10^{-3}
VI	100	2.8×10^{-6}	303	1.4×10^{-5}
VI + PEO	30	7.0×10^{-5}	370	5.6×10^{-3}
VII	100	4.9×10^{-6}	343	1.3×10^{-4}
VII + PEO	30	1.4×10^{-5}	373	4.5×10^{-4}

X, concentration of copolymer in blend; σ_{rt} , room temperature ionic conductivity; T, the highest temperature at which conductivity σ_T was measured.

obtained in the form of white amorphous powders whereas polysulfites were obtained as thick colorless oils. The average molecular weights determined for copolymers V and VI were equal to $1.66 \times 10^5 \text{ g mol}^{-1}$ and $1.15 \times 10^4 \text{ g mol}^{-1}$ respectively.

Polymer electrolytes were prepared as thin films by casting using acetonitrile (Merck, spectroscopy grade) as a solvent and LiClO_4 (Fluka, reagent grade, 10 mol% per average monomeric unit) as a dopant. However, those prepared from IV and VI did not form films and were instead obtained in a form of powders or pastes. Ionic conductivities were measured by Impedance Spectroscopy.

RESULTS AND DISCUSSION

As can be seen from the Table 1 room temperature conductivities of polysulfones based electrolytes change in a wide range (5.9×10^{-9} – $1.4 \times 10^{-5} \text{ S cm}^{-1}$) depending on kind of polymer used. The highest ambient temperature conductivity

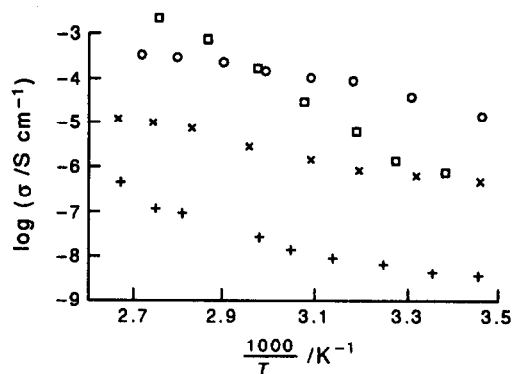


Fig. 1. Changes of conductivity *vs.* reciprocal temperature for SO_2 copolymers with various organic comonomers doped with LiClO_4 (10 mol% in respect to copolymer molecular unit concentration): +, AA- SO_2 copolymer; x, VA- SO_2 copolymer; o, AAM- SO_2 copolymer; □, $(\text{PEO})_{10}\text{LiClO}_4$ electrolyte.

(exceeding $10^{-5} \text{ S cm}^{-1}$) is measured for electrolytes containing random SO_2 -acrylamide copolymer (29% mol of SO_2 monomeric units) (V) and polysulfone having oxyethylene monomeric units in a side chain (III). The ion movements in the electrolyte III are probably governed by interactions with side chain and the conductivities found are similar to those reported for comb-like electrolytes with side chains poly (ethylene glycol)s attached to various backbones[8, 9]. In other systems oxygen atoms from both

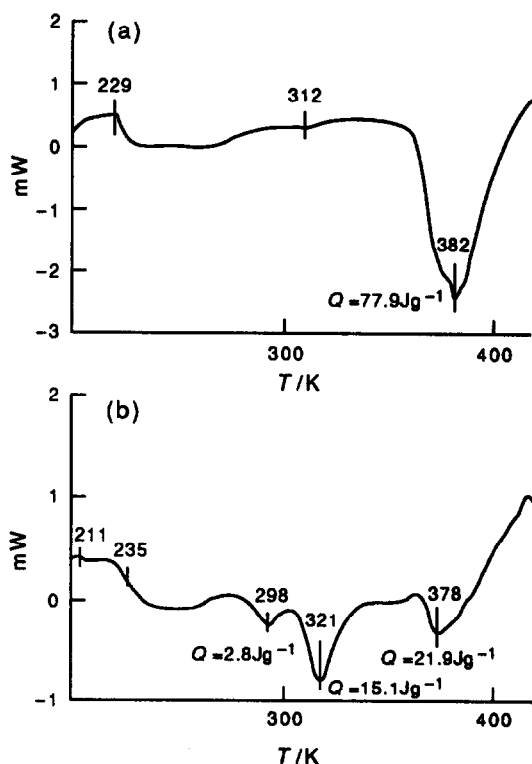


Fig. 2. DSC curves measured for (a) VII- LiClO_4 electrolyte, (b) VII:PEO = (3:7 weight ratio)- LiClO_4 electrolyte (concentration of LiClO_4 is equal to 10 mol% in respect to average monomeric units).

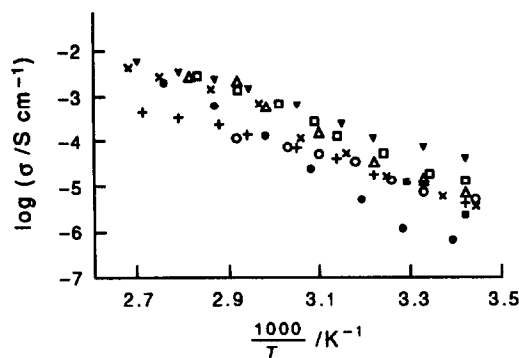


Fig. 3. Changes of ionic conductivity vs. reciprocal temperature for: ○, VII + LiClO₄; ×, VII:PEO = 2:8 + LiClO₄; □, VII:PEO = 3:7 + LiClO₄; △, VII:PEO = 4:6 + LiClO₄; +, VI:PEO = 2:8 + LiClO₄; ▽, VI:PEO = 3:7 + LiClO₄; ●, PEO–LiClO₄ electrolyte; ■, VI + LiClO₄. Concentration of polymers in blends in weight ratios. (Concentration of LiClO₄ is equal to 10 mol% in respect to average monomeric units.)

sulfonyl and side groups might be involved in ionic transport. The lowest conductivity found for electrolytes based on allyl alcohol copolymer probably results from the formation of hydrogen bonds between —OH and sulfonyl group. Higher donicity of oxygen atom in amide group than in ester ones is probably the reason of higher conductivities obtained for electrolyte V than for electrolytes II and IV. As can be seen from the results of Table 1 and Fig. 1 the activation energy for conduction in systems containing polysulfones is rather low ($E_a = 0.3\text{--}0.4$ eV) and at temperatures exceeding melting point of PEO their conductivities are lower than that measured for PEO–LiClO₄ electrolyte used as a reference material.

The ambient temperature conductivities measured for electrolytes based on polysulfites are within the range $10^{-6}\text{--}10^{-5}$ S cm⁻¹. Electrolytes based on VII copolymer were subjected to DSC experiments. DSC spectra of undoped EO–SO₂ copolymer evidences the occurrence of glass transition at approximately 210–215 K (Fig. 2). There is no trace of any crystalline melting peaks. After addition of LiClO₄ the strong melting peak ($Q_m = 77.9$ J g⁻¹) appears. It should be attributed to the presence of crystalline complex formed between polysulfite and a salt. Glass transition temperature shifts to higher temperature range ($T_g = 229$ K) which indicates interactions between copolymer and LiClO₄ also in an amorphous phase. However, this phase seems to be more flexible than in pristine PEO–LiClO₄ system for which $T_g = 245\text{--}250$ K. In spite of the presence of crystalline phase polysulfite electrolytes start to flow at relatively low temperature (308 K for VI and 345 K for VII) during the impedance experiments. Similar behaviour is also observed for electrolytes based on polysulfones (III and IV).

In order to improve the ease of processing of the copolymer, electrolytes were mixed with high molecular weight PEO ($M_w = 5 \times 10^6$). Thin flexible foils thermally stable up to 373 K were formed for blends containing more than 50 wt% of PEO. In Fig. 3 the values of conductivity measured for the (VIII)– and (VI)–LiClO₄ systems are compared with results ob-

tained for blends of these electrolytes with PEO. As can be seen, blend-based electrolytes are characterized by wider temperature stability range. The obtained conductivities are usually higher so the addition of PEO improved ionic transport.

It is well known that in many systems blending and cosolubilization of PEO with polar polymers leads to a suppression of crystallinity of the electrolytes and hence to an increase of their conductivities. DSC curve registered for PEO (70 wt%)–VII (30 wt%)–LiClO₄ blend-based electrolyte showed that the amount of crystalline PEO phase was equal to 10.1%. The formation of crystalline PEO–LiClO₄ complex phase ($T_m = 420\text{--}430$ K) is not observed (Fig. 2b). It means that polysulfite is a very effective inhibitor of crystallization of PEO. On the other hand PEO does not prevent formation of the crystalline complex between VII and LiClO₄ ($T_m = 378$ K). Small melting peak at 298 K is also found. It might be connected with interactions between polar sulfite chains and ether groups. In the low temperature range, glass transition temperatures at 211 K and 235 K are evidenced which may suggest presence of amorphous phases of different flexibility in the studied electrolytes. Low degree of crystallinity and a presence of a highly flexible polymer phase seem to be responsible for high ambient temperature conductivity of the electrolytes.

The highest room temperature conductivity 7×10^{-5} S cm⁻¹ was measured for the blend containing 30% weight of the alternating copolymer—VI. Results included in Table 1 evidenced that addition of PEO to other copolymer electrolyte systems increased their room temperature ionic conductivity which was also much higher than that of pristine PEO–LiClO₄ complex.

CONCLUSIONS

Copolymers of sulfur dioxide with vinyl monomers or ethylene oxide can be used as immobilized polymer solvents in polymeric electrolytes. Some of these exhibit conductivities exceeding 10^{-5} S cm⁻¹ at room temperature. Addition of high molecular weight PEO causes an increase of conductivity, mostly due to decreasing electrolyte crystallinity, and extends its temperature stability range. Highly polar copolymers having sulfonic or sulfinic monomeric units might find some practical application as components capable of increasing the dielectric constants of highly conductive polymeric electrolytes.

Acknowledgement—This work was partially supported by the Polish Ministry of Education within the 500/164/136/0 research programme.

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